

L2 - 13.6 MalBuster

Unknown binaries can pose a significant challenge for SOC teams, making malware reverse engineering a critical step in incident response. This hands-on investigation focused on performing **static malware analysis** to identify malicious behavior, extract indicators of compromise (IOCs), and provide actionable intelligence for threat remediation.

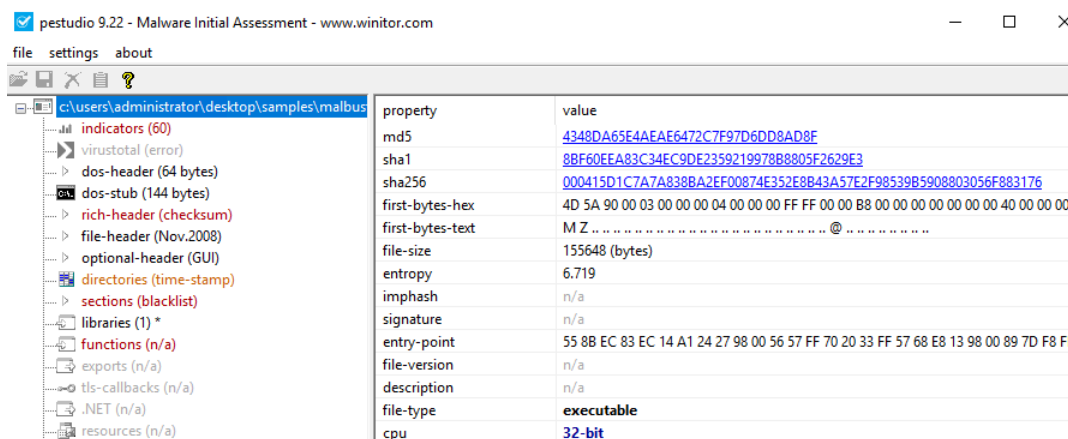
Using dedicated analysis environments such as **FLARE VM** and **REMnux**, malware samples were examined by analyzing PE headers, file characteristics, and embedded artifacts without executing the binaries. The objective was to support SOC investigations by determining the malware's functionality and uncovering evidence that could help contain and remediate the threat.

Answer the questions below

Based on the ARCHITECTURE of the binary, is malbuster_1 a 32-bit or a 64-bit application?

32-bit

Run the query: `pe-tree malbuster_1`. The **Architecture** field in **File_Header** shows **I386 (PE)**, which indicates **32-bit Windows** (whereas 64-bit shows **0x8664**).



property	value
md5	4348DA65E4AE6472C7F97D6DD8AD8E
sha1	8BF60EFA83C34EC9DE2359219978B8805F2629E3
sha256	000415D1C7A7A838BA2EF00874E352E8B43A57E2F98539B5908803056F883176
first-bytes-hex	4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 40 00 00 00
first-bytes-text	M Z @
file-size	155648 (bytes)
entropy	6.719
imphash	n/a
signature	n/a
entry-point	55 8B EC 83 EC 14 A1 24 27 98 00 56 57 FF 70 20 33 FF 57 68 E8 13 98 00 89 7D F8 F1
file-version	n/a
description	n/a
file-type	executable
cpu	32-bit

What is the MD5 hash of malbuster_1?

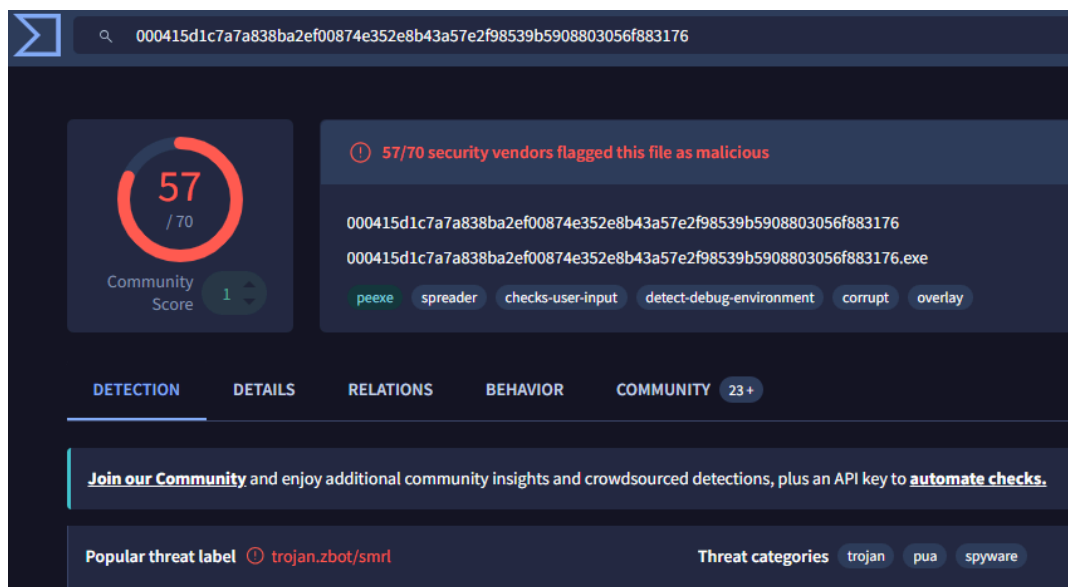
4348da65e4aeae6472c7f97d6dd8ad8f

```
C:\Users\Administrator\Desktop\Samples>ls
malbuster_1  malbuster_2  malbuster_3  malbuster_4

C:\Users\Administrator\Desktop\Samples>certutil -hashfile mal
buster_1
MD5 MD5 hash of malbuster_1: 4348da65e4aeae6472c7f97d6dd8ad8
f
CertUtil: -hashfile command completed successfully.
```

Using the hash, what is the popular threat label of malbuster_1 according to VirusTotal?

trojan.zbot/razy

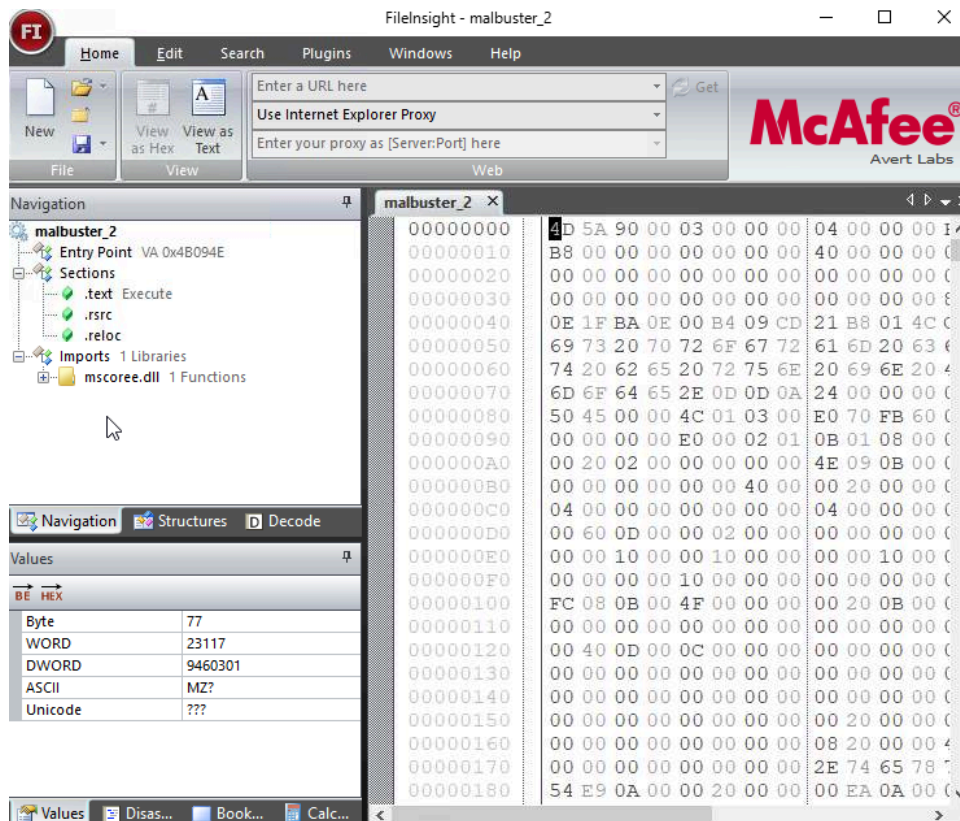


Based on VirusTotal detection, what is the malware signature of malbuster_2 according to Avira?

HEUR/AGEN.1306860

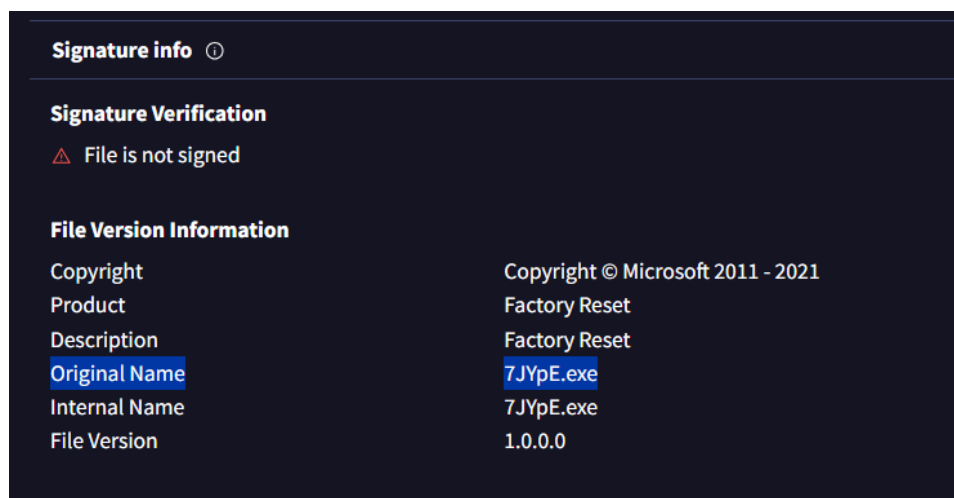
malbuster_2 imports the function _CorExeMain. From which DLL file does it import this function?

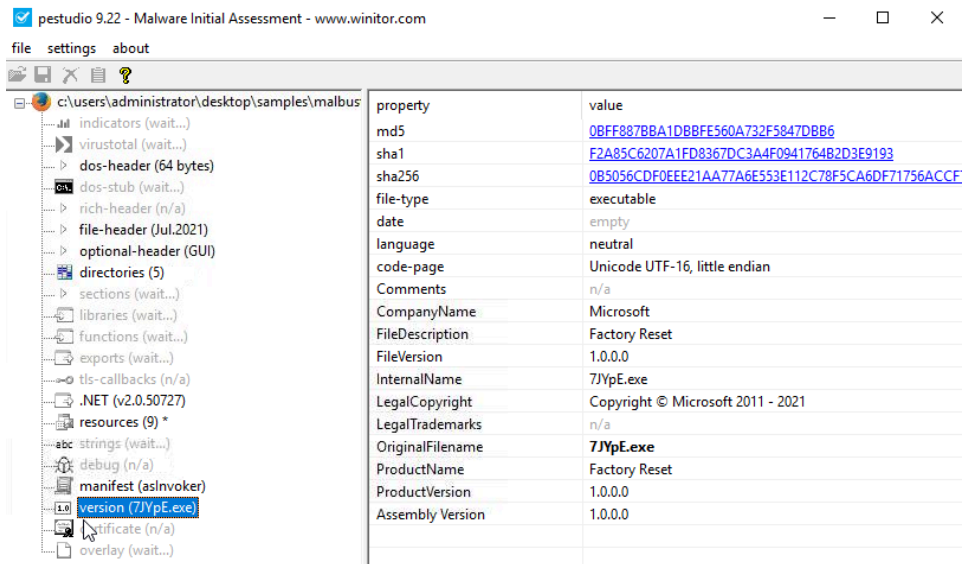
mscoree.dll



Based on the VS_VERSION_INFO header, what is the original name of malbuster_2?

7JYpE.exe





Using the hash of malbuster_3, what is its malware signature based on abuse.ch?

TrickBot

Browse Database

See search syntax see below, example: tag:TrickBot Search

Search Syntax ?

Search:

Date (UTC)	SHA256 hash	Type	Signature	Tags	Reporter	DL
2021-07-22 17:12	9da8a5a0b5957db6112e...	dll	TrickBot	dll rob109 TrickBot	abuse_ch	

Using the hash of malbuster_4, what is its malware signature based on abuse.ch?

ZLoader

Browse Database

See search syntax see below, example: tag:TrickBot

Search

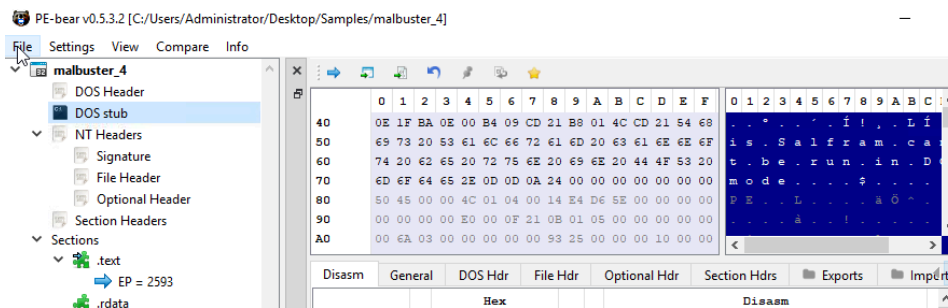
Search Syntax ?

Search:

Date (UTC)	SHA256 hash	Type	Signature	Tags	Reporter	DL
2020-07-05 20:48	00272dd639402fa76db4...	exe	ZLoader	dll ZLoader	Racco42	

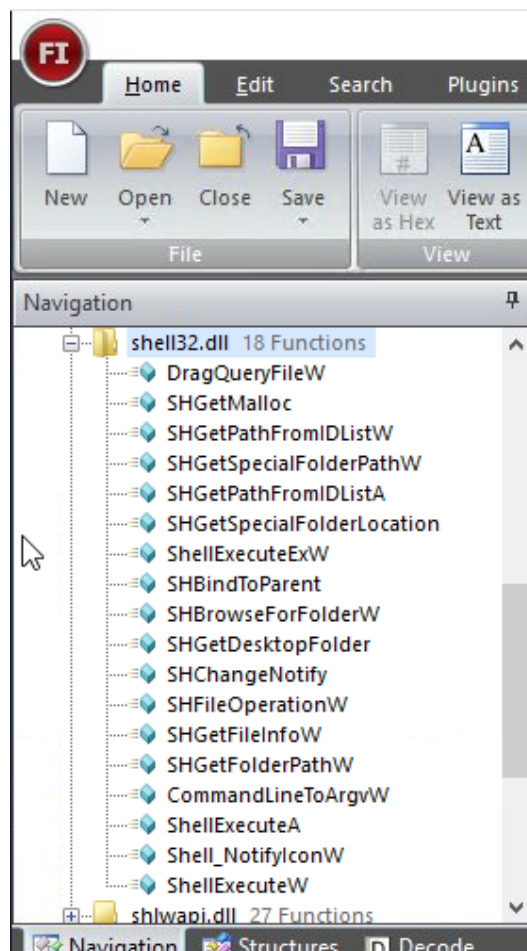
What is the message found in the DOS_STUB of malbuster_4?

!This Salfram cannot be run in DOS mode.



malbuster_4 imports the function ShellExecuteA. From which DLL file does it import this function?

shell32.dll



Using capa, how many anti-VM instructions were identified in malbuster_1?

3 (Run the command `capa.exe Samples\malbuster_1`, then check how many **anti-VM rules** were triggered in the results.)

Using capa, which binary can log keystrokes?

malbuster_3 (run the command : `capa malbuster_*` The capa output shows that malbuster_3 matches keystroke-logging capabilities.)

Using capa, what is the MITRE ID of the DISCOVERY technique used by malbuster_4?

T1083 (run the command: `capa malbuster_4`, check the discovery section)

Which binary contains the string GodMode?

malbuster_2 (Run the command `strings malbuster_* | grep GodMode` shows that **malbuster_2** contains the string.)

Which binary contains the string Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)?

malbuster_1 (run the command: `strings malbuster_* | grep "Mozilla/4.0"`)